BICOL CACAO PITCH · SPEAKER SCRIPT

Keisuke

Tradeoffs / philosophy voice

How to use this: not a verbatim script — nobody is expected to read from it word-for-word, and realistically no one will stick to a script anyway. Study it well enough to explain each point in your own words, improvise if you're cut off, and answer a follow-up without freezing. If a question lands outside your section, give the short version and hand off by name — every presenter's script and the underlying `FIELD-NOTES.md` are in the same GitHub repo.

日本語 (JAPANESE)

あなたはトレードオフ・哲学担当です。対象範囲の設定と制約条件があなたの担当スライドに含まれています。

Field-visit recap (read this first if you missed a visit)

Everything in the deck traces back to these visits — this is the shared ground truth every presenter should know, whether or not you were there.

Tatay's cacao farm, Batbat (Tuesday, Aug 26). A 2.5-hectare farm run by Tatay alone — his children work in the cities — 100% organic, and he has health problems that limit how much heavy labor he can do himself. His farm used to produce about 500 kg of cacao a year (sold at ₱75–100/kg); this season it's close to **zero**, because of pests and disease. His own estimate of the farm's potential if pests were solved: **2,000 kg/year** — a 4× recovery, the biggest single lever we found anywhere. Cacao flowers year-round here, so the pest pressure never gets an off-season. On water, his framing was specific: the constraint is water *pressure*, not volume — he can't even estimate how many sprinklers his source could feed — and he's open to drip irrigation and rainwater collection. Important attribution point: Tatay described the problems and constraints; the pilot plan, the tools, and every proposed solution came from our own investigations, not from him.

What the farmers and Muravah told us directly. Pest control is their #1 problem — ahead of typhoon, which everyone called "unpredictable" and would rather we not chase. Post-harvest losses run 80–100% from pests/disease across the network, and 60% of pods die from careless handling alone (worse under Mayon ashfall). Farmer income is ₱2,000–10,000/month — below Bicol's ₱13,624/month poverty line for a family of five — the average farmer is about 50, and there's no succession pipeline; Muravah copes by recruiting BU Guinubatan students. Two hard rules for any solution: **organic only** (no synthetic fertilizers or pesticides, ever) and **no AI/robotics** ("not there yet," "AI is biased according to the natives") — low-tech and human-in-the-loop is a values position for them, not a budget issue. More labor doesn't raise profit, because pest pressure caps yield no matter how much work goes in.

Muravah Foundation / Mayon Gold. The cacao NGO anchoring all of this: cacao profits fund typhoon-proof homes and scholarships, heirloom native cacao interplanted with Pili and coconut, and a production-cycle data study already running across 500–1,000 farmers — which is where our pilot would live.

The processing facility. Their own #1 problem is power outages, but the line that matters for us: they're "sometimes out of supply for cacao." Farm-level pest losses upstream directly destabilize the factory downstream — fixing the farms fixes two problems at once.

The other farms (Aug 27–28, now Extras slides). Tatay Aldavis's rice/dairy farm school (a different "Tatay" — it's a Bikol honorific for an elder, not a name) controls pests with herbal sprays and ducks that eat snails — proof organic pest control works in Bicol; plus a bamboo farm, stingless-bee honey producers, the Zambales agrivoltaics site (the ₱10M funding benchmark), and a youth agritourism program.

Your slides

You're the tradeoffs/philosophy voice — scope and constraints sit in your slides.

Slide 4 — Two fixable problems, not every problem (scope).

This is the honesty slide — we're not claiming to fix everything. **Out of scope:** typhoon and Mayon ashfall. Every source called typhoon "unpredictable" and said they'd rather see pests and disease fixed; we also leave alone what's already working — waste, sanitation, land tenure, transport. **In scope:** flood/drought resilience plus pest and disease control — fixable, with a **4× yield recovery** sitting on the table. Close with the design rule that governs the rest of the deck: *adapt, don't invent* — every solution ahead already works on another Bicol farm.

日本語 (JAPANESE)

これは誠実さを示すスライドです。私たちはすべての問題を解決すると主張しているわけではありません。対象外:台風とマヨン山の降灰。すべての情報源が台風を「予測不能」と呼び、それよりも病害虫を解決してほしいと明言しました。すでにうまくいっている分野(廃棄物処理、衛生、土地制度、輸送)にも手を付けません。対象:洪水・干ばつへの耐性と、病害虫の防除です。これは解決可能で、4倍の収量回復という成果がすぐ手の届くところにあります。最後に、このプレゼン全体を支配する設計方針で締めてください:「発明ではなく、応用する」。この後に提案するすべての解決策は、ビコールの別の農場ですでに機能しているものです。

Slide 5 — More constraints we have to design around.

1. Muravah has **no annual drought/flood prediction**, so they sync planting to the monsoon and specifically seek out farmers who already have water access — a real infrastructure gap.
2. The farmer population is aging (~50 average) with no government training pipeline; Muravah's own workaround is recruiting students from BU Guinubatan.

3. Hard constraint: **no synthetic fertilizers or pesticides**, biopesticides/biofertilizers only — this rules out entire categories of "obvious" agri-tech fixes before we even start.
4. Muravah is explicitly skeptical of AI and robotics — quote it directly: "not there yet, other problems to solve," and "AI is biased according to the natives." Frame this as a values position, not just a budget constraint — it's why every solution later in the deck is deliberately low-tech and human-in-the-loop.

日本語 (JAPANESE)

1. ムラワ財団には干ばつや洪水を年単位で予測する仕組みがないため、モンスーンの時期に合わせて作付けを行い、すでに水源を確保している農家を意図的に探しています。これは実質的なインフラの欠如です。 2. 農家の高齢化が進んでおり(平均年齢約50歳)、政府による後継者育成の仕組みはありません。ムラワ財団自身の対応策は、BUギヌバタン(ビコール大学ギヌバタン校)の学生をリクルートすることです。 3. 譲れない制約条件として、合成肥料や合成農薬は使用せず、生物由来の防除・肥料のみを用います。これにより、一見「わかりやすい」農業テック的な解決策の多くは、検討の入り口の時点で除外されます。 4. ムラワ財団はAIやロボット技術に対して明確に懐疑的です。「まだその段階ではない、他に解決すべき問題がある」「AIは先住民の視点からすると偏っている」という発言をそのまま引用してください。これは単なる予算上の制約ではなく、価値観に基づく立場として伝えてください。このプレゼンで後に紹介するすべての解決策があえてローテクで、人が関わる仕組みになっている理由です。

Slide e4 — Vertical panels waste potential output (Extras).

Short, standalone point — don't over-explain it, let it land. Mounting solar panels vertically is not the efficient way to collect sunlight; a wall-mounted panel captures noticeably less irradiance than an optimally tilted one (per Masa's chart on e3: about 65% vs. 100%). Be upfront that this is **our own engineering observation, not something a source told us in an interview** — a design point worth carrying into any agrivoltaic or solar-irrigation proposal built on the Zambales precedent. The two photos are our own shots of the vertical panels at the site, front and back.

日本語 (JAPANESE)

短く、独立した論点です。説明しすぎず、そのまま印象に残す程度で構いません。太陽光パネルを垂直に設置することは、太陽光を効率的に取り込む方法ではありません。壁面設置のパネルは、最適な角度に傾けたパネルに比べて明らかに受光量が少なくなります(マサのe3のグラフによると、約65%対100%です)。これはインタビューで聞いた内容ではなく、私たちチーム自身の工学的な見解であることを率直に伝えてください。ザンバレスの事例をもとにした太陽光農業(アグリボルタイクス)や太陽光灌漑の提案において、留意すべき設計上のポイントです。スライドの2枚の写真は、現地で私たち自身が撮影した垂直設置パネルの正面と背面です。

Slide e5 — Solutions we probed and gathered from other farms (Extras).

This is the payoff slide for "adapt, don't invent." Of 12 candidate pest/disease solutions brainstormed, only **6 survived independent fact-checking** against the actual interview data — say that filtering process out loud, it's a credibility point. Three groups:

- *Monitoring & routine discipline (3)*: a cacao-adapted 15-day treatment calendar borrowed from the taro project's proven schedule; a field-officer symptom-logging checklist; a centralized compliance-and-bio-input ledger.
- *Biological & cultural control (2)*: an herbal foliar/pod spray adapted from the rice/dairy farm's practice; an amino-acid/molasses bio-concoction on the taro project's 15-day cycle.
- *Post-harvest handling (1)*: rolling phytosanitation rounds every 7–14 days.

Land the recommendation clearly: for Tatay specifically, start with the **15-day calendar plus the herbal spray** — cheapest to start, no new equipment, fits his limited labor capacity given his health constraints. The ledger and the amino-acid spray follow once the calendar habit is established.

日本語 (JAPANESE)

このスライドは「発明するのではなく、応用する」という考え方の集大成です。ブレインストーミングで出た12件の病害虫対策候補のうち、実際のインタビューデータと照らし合わせた独立した事実確認を経て残ったのはわずか6件です。この絞り込みのプロセスを口頭できちんと伝えてください、信頼性を示す重要なポイントです。3つのグループに分かれます。 モニタリングと定期的な規律の徹底(3件):タロイモプロジェクトで実績のあるスケジュールを応用した、カカオ向け15日サイクルの防除カレンダー。現場担当者による症状記録チェックリスト。防除の実施状況と生物由来資材の使用を一元管理する台帳。 生物学的・文化的防除(2件):米・酪農農場の実践を応用した、葉面・ポッド用のハーブ系スプレー。タロイモプロジェクトの15日サイクルを応用した、アミノ酸・糖蜜による自家製防除液。 収穫後の取り扱い(1件):7〜14日ごとに行う、巡回による衛生管理(収穫と病害ポッドの除去・焼却を一度に行う)。推奨事項ははっきりと伝えてください。タタイさんの場合は、まず15日サイクルのカレンダーとハーブ系スプレーから始めるのが最も現実的です。初期費用が最も安く、新たな設備も不要で、健康上の制約がある彼の限られた労働力にも合っています。台帳管理とアミノ酸スプレーは、カレンダーの習慣が定着してから導入します。